Srihith Sai Kurelli

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.16
Fiitjee Junior College	Class XII (Telangana Board of Intermediate Education)	2024	97.70%
Delhi Public School Nacharam	Class X (CBSE)	2022	98.80%

SCHOLASTIC ACHIEVEMENTS

- Achieved **International Rank 3** in the **SOF International Mathematics Olympiad (IMO)**. **(Grade 10)**
- Awarded **High Distinction** in the **International Chemistry Quiz (ICQ)** by the Royal Australian Chemical Institute (RACI). **(Grade 10)**
- Awarded **Gold Honour** in the **International Youth Math Challenge (IYMC)**, placing in the top 3% worldwide. **(Grade 12)**
- Attained an **SGPA** of **9.72** in **Semester 2** through consistent academic excellence.

PROJECTS

Story Completion using LSTM | *Python, TensorFlow, Keras, Git, NLP* | *GitHub Link*

- Engineered an end-to-end LSTM-based NLP pipeline for story completion, encompassing text preprocessing, tokenization, sequence generation, model training, and evaluation.
- Generated **644K+ supervised training sequences** from the TinyStories dataset using a **12K-word vocabulary** and sliding windows.
- Implemented **Greedy, Temperature, and Top-k decoding** strategies to generate coherent story continuations while evaluating the trade-off between text coherence and diversity.

Codeforces Problem Recommender | *Python, XGBoost, Scikit-Learn, Codeforces API, Git* | *GitHub Link*

- Built an end-to-end machine learning recommendation system by collecting **Codeforces API** data, engineering numerical and tag-based features, and constructing a supervised dataset with **212K+ user-problem interactions** spanning **10K+ problems**.
- Compared Logistic Regression, Random Forest, and XGBoost models, achieving a **ROC-AUC of 0.876** with XGBoost, and implemented a personalized recommendation engine that ranks unseen problems based on predicted solve likelihood.

Credit Card Fraud Detection | *Python, XGBoost, Scikit-Learn, Pandas, Git* | *GitHub Link*

- Developed a machine learning model to classify fraudulent and genuine credit card transactions using **XGBoost**, addressing severe class imbalance through weighted learning and evaluating model performance using multiple classification metrics.
- Performed **data analysis**, preprocessing, and model serialization with **Joblib**, achieving **99.93% Accuracy** and **80.79% F1-Score**.

URL Shortener API | *Python, FastAPI, PostgreSQL, SQLAlchemy, Git* | *GitHub Link*

- Developed a RESTful URL Shortener API using **FastAPI**, **PostgreSQL**, and **SQLAlchemy**, enabling unique short URL generation, seamless redirection, and persistent storage with environment-based configuration.
- Implemented click analytics with usage tracking, last-accessed timestamps, **Pydantic** request validation, and documented REST endpoints using **Swagger UI** and **Postman** for API testing.

Two-Process Othello Game | *C++, POSIX IPC, SDL2, Git* | *GitHub Link*

- Developed a two-process implementation of the Othello board game using **POSIX shared memory** (shm_open, mmap) and **named semaphores** for synchronized communication and consistent game-state updates between independent client and server processes.
- Built an **SDL2-based graphical client** supporting mouse-driven gameplay, legal move highlighting, undo/redo functionality, and automatic restoration of saved game state through binary persistence.
- Designed a modular C++ application with separate modules for game logic, IPC, persistence, rendering, and **RAII wrappers**.

SKILLS

Programming Languages : C, C++, Python, SQL

Core Concepts : Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

Machine Learning : Scikit-learn, TensorFlow, Keras, XGBoost

Python Libraries : NumPy, Pandas, Matplotlib, Seaborn

Backend & Databases : FastAPI, PostgreSQL, SQLite

Developer Tools : Git, GitHub, Postman, VS Code, Jupyter Notebook

Soft Skills : Critical Thinking, Communication, Leadership, Collaboration

RELEVANT COURSES

- **Computer Science :** Introduction to Programming, Data Structures and Applications, Algorithms, Theory of Computation, Discrete Mathematics for Computer Science, Introduction to Computing, Operating Systems-I
- **Artificial Intelligence & Machine Learning :** Introduction to AI, Machine Learning Specialization (DeepLearning.AI)
- **Mathematics :** Probability and Random Variables, Introduction to Statistics, Elementary Linear Algebra, Differential Equations, Complex Variables, Calculus I & II

POSITIONS OF RESPONSIBILITY

Head of Operations | Entrepreneurship Cell, IIT Hyderabad

- Leading a team of 10 Managers and 20+ Assistant Managers for the successful execution of E-Summit and E-Merge.
- Overseeing logistics, vendor negotiations, and speaker hospitality for flagship E-Cell events.

UG Mentor | Sunshine (Counselling Cell), IIT Hyderabad

- Mentored 28 first-year students, supporting their academic, personal, and social transition to campus life.
- Actively volunteered in mental well-being initiatives and awareness events, promoting student wellness.

Ideation Coordinator | Finance & Consulting Club, IIT Hyderabad

- Conceptualized and designed engaging events for the club's flagship event, HORIZON'26.
- Identified speakers, curated session content, and managed the execution of events like Campus Heist.

EXTRACURRICULAR

- Volunteered with **Nirmaan Organization's Career Saathi Program**, mentoring students and developing career-readiness content.
- Solved **300+ LeetCode problems**, strengthening problem-solving skills and proficiency in data structures and algorithms.
- Volunteered with the **National Service Scheme (NSS)** at **IIT Hyderabad**, promoting **community service** through

social outreach.